

Why df-hT shows fuseblk but lsblk-f shows ntfs

1. lsblk -f → Shows actual filesystem on disk

- lsblk -f reads **filesystem metadata directly from the block device**
- It tells you **what filesystem is actually created on the USB**

So it shows: ntfs

✓ Because your USB (/dev/sdb) is formatted with **NTFS**

✓ This is the **real on-disk filesystem**

2. df -hT → Shows how filesystem is mounted

- df -hT shows the **filesystem as handled by the OS (kernel/mount layer)**
- It reflects **how Linux is accessing the filesystem**

So it shows: fuseblk

Why fuseblk?

- Linux does **not fully rely on native kernel drivers for NTFS (in many setups)**
- Instead, it uses **FUSE (Filesystem in Userspace)**

Therefore:

- NTFS is mounted via FUSE
 - Appears as **fuseblk (FUSE block device)**
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What is fuseblk?

fuseblk = Filesystem mounted using FUSE as a block device

What is FUSE?

- **FUSE (Filesystem in Userspace)** allows filesystems to run in **user space instead of kernel space**
 - Features:
 - Safer (less risk to kernel)
 - Flexible
 - Slightly slower than kernel drivers
 - Runs in user space (not kernel space)
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Special Case (Modern Systems)

- Newer Linux kernels support **native NTFS driver (ntfs3)**

In that case, `df -hT` may show: **ntfs3**

